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A Statistical Audit of LLM Calibration Heterogeneity Across Knowledge Domains

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└ What is calibration — and why does it matter?

[45 sec]

Open with the definition — make sure the audience knows what calibration means before anything else. The formula is on the slide so you don't need to dwell on it; just say “ECE is the weighted average gap between confidence and accuracy across confidence bins.”

The key move is the fairness analogy: just like we'd never accept a single accuracy number as a fairness audit, a single ECE number is not a calibration audit. Land that point before moving on — it's the whole motivation.

If anyone in the room works on ML deployment, you can add: “when a model says it's 90% confident, users treat that as a reliability signal — if that signal is only accurate on average, users in certain domains are being systematically misled.”

[If asked: how does the ECE formula work?] You divide the confidence range $[0, 1]$ into equal-width bins. Every question is assigned to a bin based on its confidence value — e.g. a question where the model is 87% confident goes into the 0.8–0.9 bin. B_b is the set of *all* questions whose confidence falls in bin b , regardless of whether they were correct. $\text{acc}(B_b)$ is the fraction of those questions the model got right. $\text{conf}(B_b)$ is the simple average of confidence values for all questions in that bin. The comparison is: of all questions where the model said it was $\sim 85\%$

What is calibration — and why does it matter?

LLMs are deployed across diverse domains — law, medicine, mathematics, ethics. Users treat model confidence as a reliability signal.

A model is **well-calibrated** if its stated confidence tracks empirical accuracy:

- 80% confident $\Rightarrow$ correct 80% of the time

Miscalibration is typically reported as a **single number** (Expected Calibration Error):

$$\text{ECE} = \sum_b \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)|$$

The problem: a single aggregate ECE hides where and how much miscalibration occurs across different domains.

Analogy: reporting one accuracy number across demographic groups is insufficient for a fairness audit. The same logic applies to calibration.

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Research questions

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Setting: MMLU benchmark — 14,042 questions, 57 subjects, 4 domains (STEM, Humanities, Social Sciences, Other) — queried with GPT-4o-mini and Llama-3-8B-Instruct.

- How much of calibration variance is attributable to domain vs. subject vs. individual question?
- What question features predict the direction and magnitude of miscalibration?
- Are patterns consistent across models, or model-specific?

Framework

Three-level mixed model + isotonic calibration curves + empirical Bayes shrinkage + Benjamini-Hochberg FDR correction

[30 sec]

Read the three questions quickly — the audience will follow. For the framework block: “I won’t spend time on each method now; the key idea is that we want statistically rigorous answers, not just descriptive bar charts.”

Don’t explain what BH correction is here — that comes up naturally in the robustness slide. Just name-drop it so the audience knows it’s coming.

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└ Data & confidence elicitation

Data & confidence elicitation

MMLU: 14,042 four-choice questions, 57 subjects, 4 domains (questions → subjects → domains)

Two models:

- GPT-4o-mini (OpenAI API, top_logprobs=20)
- Llama-3.8B-Instruct (Colab T4, 4-bit quantization)

Confidence measure → renormalized softmax over the four answer tokens:

$$c_i = \max_{a \in \{A, B, C, D\}} \frac{e^{g^i_{a,i}}}{\sum_{a \in \{A, B, C, D\}} e^{g^i_{a,i}}}$$

Primary outcome → signed calibration gap:

$$\text{gap}_i = c_i - y_i \quad (y_i \in \{0, 1\})$$

[45 sec]

Explain the hierarchy briefly: “MMLU has 57 subjects like clinical medicine, formal logic, high school math — grouped into 4 broad domains. That nested structure is what makes a multilevel model the right tool.”

For the confidence measure: “We extract the log probability the model assigns to each of the four answer tokens, renormalize them to sum to 1, and take the max. This is a standard approach — it gives us a proper probability on the model’s chosen answer.”

For the signed gap: “The nice thing about using the signed gap instead of ECE is that it tells us the *direction* — positive means the model thinks it knows more than it does.”

One methodological note if it comes up: we had to use top_logprobs=20 for GPT because with 5 logprobs, 55% of questions had at least one answer token missing, which would artificially inflate confidence estimates. That’s a real data quality issue that took some debugging.

[Reference: gap vs. ECE — what is used where]

- **Calibration gap** (question-level, signed): outcome for the multilevel regression; BH t-tests test whether subject mean gap = 0; bar chart shows shrunken subject mean gaps (BMP)

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Hierarchical model

Hierarchical model	
Three-level mixed model	
Questions (i) nested in subjects (j) nested in domains (k):	
$\text{BAP}_{ijk} = \mu + \mathbf{x}_{ijk}^T \beta + u_k + v_j + \varepsilon_{ik}$	
Fixed effects	Random effects
μ : global intercept (mean overconfidence)	$u_k \sim \mathcal{N}(0, \sigma_{\text{domain}}^2)$
β : question-level covariates	$v_j \sim \mathcal{N}(0, \sigma_{\text{subject}}^2)$
(word count, entropy, negation, max choice length) — all standardized	$\varepsilon_{ik} \sim \mathcal{N}(0, \sigma^2)$
	Estimated via REML (<code>estatmodel.remlsd30</code>)
ICC decomposition: what fraction of calibration gap variance is between-subject vs. within-subject?	

[30 sec]

Walk through the model briefly: “We model the signed calibration gap — confidence minus accuracy — at the question level, with random intercepts for both subject and domain.”

Emphasize the key question: “The ICC decomposition will tell us whether the miscalibration is driven by specific subjects, or whether it’s spread evenly across all questions regardless of subject.”

Mention REML: “We use restricted maximum likelihood, which gives unbiased variance component estimates — important here since the variance decomposition is our main inferential target.”

[If asked: why REML instead of standard MLE?] Both maximize a likelihood over the variance components, but they differ in *what* likelihood they maximize.

MLE treats the fixed effect estimates $\hat{\mu}, \hat{\beta}$ as known with certainty when computing residuals — as if no degrees of freedom were spent estimating them. This causes MLE to systematically *underestimate* variance components, analogous to dividing by n instead of $n - p$ when computing a sample variance.

REML marginalizes out the fixed effects first and maximizes the likelihood of a set of contrasts

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└ Finding 1: heterogeneity is real but concentrated at the question level

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RQ1: How much of calibration variance is attributable to domain vs. subject vs. question?

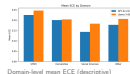
Table: Variance decomposition (both models)

Level	GPT-4o-mini		Llama-3.8B	
	Var	ICC	Var	ICC
Domain	0.000	0.000	0.000	0.000
Subject	0.013	0.079	0.004	0.020
Residual	0.150	0.922	0.190	0.980

Both models: between-subject structure exists but is modest.

GPT: 7.9% between-subject ICC.

Llama: 2.0% — more diffuse miscalibration.



[30 sec]

Lead with the structure: “Both models show that the vast majority of calibration variance — 92 to 98% — is at the question level. But a meaningful slice is systematically concentrated at the subject level. That’s not noise — it’s structure.”

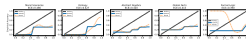
Highlight the difference: “GPT shows 7.9% between-subject ICC, Llama shows 2.0%. Llama’s miscalibration is more diffuse — spread more evenly across subjects rather than concentrated in particular domains.”

[Important: what ICC does and does not say] ICC = Intraclass Correlation Coefficient = $\sigma^2_{\text{subject}} / \text{total variance}$. Intuition: if you drew two questions at random from the *same* subject, ICC is how correlated their calibration gaps would be due to shared subject membership. ICC = 0 means subject tells you nothing; ICC = 1 means every question in a subject has the same gap. ICC = 7.9% does NOT mean calibration barely varies across subjects. It means 7.9% of total question-level variance is attributable to subject membership. That 7.9% is still large enough to produce a 13× range in mean subject gaps (0.037 to 0.487) — the two facts are compatible. Most question-to-question variability is noise regardless of subject (the 92%), but the systematic component that does exist is substantively meaningful

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└ Finding 2: all 57 subjects are overconfident on average

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RQ1: Visual evidence of subject-level miscalibration pattern



The 5 most miscalibrated subjects shown. All 57 subjects have positive mean calibration gap. In the high-confidence region where data is concentrated, curves lie below the diagonal throughout. Isotonic (solid) and kernel-smoothed (dashed) curves agree where support exists; divergence at low confidence reflects empty bins.

[30 sec]

Lead with the universal claim carefully: “All 57 subjects have a positive mean calibration gap — they are overconfident on average. These 5 curves show what that looks like in the region where data is actually concentrated, which for most subjects is the high-confidence end.”

Be precise: the below-diagonal pattern holds in the region of support (roughly 0.85–1.0 for most subjects). It is NOT correct to say every point on every curve is below the diagonal — 23/57 subjects have parts of their isotonic curve above the diagonal, but this is at low-confidence bins where there are essentially no data points. The mean gap being positive is the correct universal claim.

On the formal_logic kernel curve specifically: the dashed curve reads near 1.0 from confidence 0.03 to 0.53. This is a smoothing artifact — formal_logic has zero questions in those bins (126 total questions, 89 of which are in the 0.95–1.00 bin). The kernel is smoothing over empty space and should not be interpreted there. The isotonic curve is more honest: it just holds the last observed value flat across empty bins.

Point out isotonic vs. kernel agreement where it matters: “In the region where data exists — above 0.7 or so — the solid and dashed lines track each other, which validates the monotonicity

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— Finding 3: universal overconfidence, 13× range in magnitude

[45 sec]

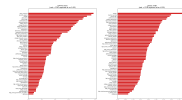
“57/57 rejected — but don't read too much into the count. RLHF fine-tuning pushes every subject above zero by default: the global intercept is 0.196, meaning there's a baseline overconfidence floor before any subject-level differences enter. With 100–1,500 questions per subject and that strong global signal, the t-tests have enough power to flag everything. BH has 57 near-zero p-values and nothing to correct.

The interesting finding isn't that all 57 reject — it's that the *magnitude* varies 13-fold. Some subjects are essentially well-calibrated; others are off by nearly 50 percentage points. That heterogeneity is invisible in a single aggregate number.”

[If asked: where does empirical Bayes / shrinkage come in?] The bars in this chart are *shrunk* subject mean gaps — BLUPs (Best Linear Unbiased Predictors) from the mixed model, not raw subject averages. Subjects have very different sample sizes (100 to 1,534 questions). A small subject with 100 questions has much noisier estimates — it could look extremely miscalibrated just by chance. Empirical Bayes shrinkage corrects for this by pulling each subject's estimate toward the grand mean (0.196), with more shrinkage for smaller subjects and less for larger ones.

Finding 3: universal overconfidence, 13× range in magnitude

RQ1: How large and heterogeneous is subject-level miscalibration?



57/57 subjects rejected after BH correction ($\alpha = 0.05$).
 Subject mean gap > 0:
 > 0: overconfident on average,
 < 0: underconfident on average.
 But the magnitude varies drastically (GPT-4o-mini):

- Max: moral scenarios (mean gap = 0.487)
- Min: marketing (mean gap = 0.037)
- Range: 13×

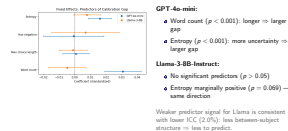
Humanities subjects cluster at the low STPM and “Yellow” at the

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What predicts the gap? Fixed effects

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RQ2: What question features predict the direction and magnitude of miscalibration?



[30 sec]

Note: these are **two separate models**, each fit independently on its own data. GPT and Llama each produce their own β estimates — the two columns on this slide are not from a single joint model.

Lead with the GPT entropy finding — it's counterintuitive: “When GPT-4o-mini spreads its probability mass across options, expressing uncertainty, the gap actually gets larger. The model isn't reliably correct more often when it's hedging; it's miscalibrated even in its expression of uncertainty.”

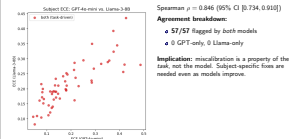
Then contrast with Llama: “Llama shows no significant predictors. The trend for entropy is in the same direction but doesn't reach significance. This is consistent with Llama having less between-subject structure overall — there's simply less systematic variance to explain.”

[If asked: why do GPT and Llama differ in fixed effect significance?] GPT's ICC is 7.9% — its miscalibration is organized by subject, concentrating in subjects like abstract algebra and moral scenarios. These subjects also tend to have longer, more ambiguous questions. So word count and entropy are picking up real signal because they correlate with the hard subjects. Llama's ICC is only 2.0% — its miscalibration is more diffuse, not concentrated in particular

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└ Cross-model agreement: miscalibration is task-driven

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RQ3: Are miscalibration patterns consistent across models, or model-specific?



[45 sec]

[Big picture: how this slide connects to earlier findings] Three separate claims have been made so far — it is easy to conflate them:

1. **Universal overconfidence** (global intercept = 0.196): all 57 subjects are overconfident because the model itself is globally biased by RLHF. The 57/57 BH rejection is mostly detecting this global effect, not between-subject differences.
2. **Between-subject heterogeneity in magnitude** (ICC = 7.9%, 13 $\times$ range): on top of the global bias, subjects vary in how badly miscalibrated they are. 7.9% of total variance is between-subject — modest, but enough to produce a 13 $\times$ range in mean gaps that matters practically.
3. **Task-driven miscalibration** (this slide, $\rho = 0.846$): whatever between-subject variation exists is consistent across two very different models. This answers a different question than ICC — not “how much variation is there?” but “where does that variation come from?”

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└ Sensitivity analysis: are findings robust?

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Three stress tests — run on both models.

1. ECE binning & NLCS

$b \in \{10, 15, 20, 25, 30\}$ bin.

GPT: $\rho \geq 0.997$

Llama: $\rho \geq 0.984$

vs. unbinned NLCS: $\rho \geq 0.861$

Rankings stable.

2. FDR threshold

BH at $\alpha \in \{0.01, 0.05, 0.10\}$.

GPT: 56/57 rejected

Llama: 55/57 rejected

Not driven by α .

3. Model structure

3-level vs. 2-level model.

GPT: $ICC_{adj} = 0.079$

Llama: $ICC_{adj} = 0.020$

(identical in both specs)

ICC robust.

[45 sec]

Frame briefly: “Before trusting these results, I stress-tested the three main modeling choices across both models.”

Column 1: “We vary the number of confidence bins used to compute ECE — 10, 15, 20, 25, 30 — and check whether subject rankings change. To be clear: ECE bins questions by their *confidence value*, not by the gap. For each bin, we compare mean accuracy to mean confidence across the questions in that bin. Changing bin count shifts absolute ECE values slightly but barely moves the subject ordering: $\rho \geq 0.997$ for GPT. We also compare against the unbinned NLCS as a completely bin-free alternative: $\rho \geq 0.861$. Our rankings aren’t artifacts of binning choices.”

Column 2: “At $\alpha=0.01$, GPT still rejects 56/57 and Llama rejects 55/57. The two Llama subjects that drop out at the strictest threshold have adjusted p-values of 0.014 and 0.031 — just above the cutoff. The finding is not alpha-sensitive.”

[If asked: what does varying the BH α tell us?] BH α controls the expected proportion of false discoveries among all rejected tests. At $\alpha = 0.05$, at most 5% of rejections are expected to be false; at $\alpha = 0.01$, at most 1%

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Takeaways

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- ❖ **The unit of analysis is the subject, not the model.** Aggregate ECE hides a $13\times$ range in miscalibration magnitude. 57/57 subjects rejected is the wrong headline — the magnitude variation is what matters.
 - ❖ **Miscalibration is task-driven, so subject-level fixes are needed.** Spearman $\rho = 0.846$ across two very different models means the same subjects are hardest regardless of architecture or training. Global temperature scaling cannot fix subject-specific structure.
 - ❖ **The audit framework generalizes.** Multilevel model + BH correction + shrinkage applies to any benchmark with a hierarchical structure — a reusable tool for honest calibration evaluation.
 - ❖ **Open questions remain.** Two models is suggestive, not definitive. We can describe which subjects are hard but not why — abstractness, training data rarity, or something else is still unknown.
- Next steps: subject-specific temperature scaling; broader cross-model comparison; extend to open-ended generation.

[45 sec]

Don't read the bullets — synthesize across all four:

“The core finding is that the subject is the right unit of analysis for calibration, not the model. Aggregate ECE hides a 13-fold range; 57/57 rejected sounds impressive but is mostly a consequence of large sample size and global overconfidence — the magnitude variation is the finding that matters.”

“The reason subject-level recalibration is needed, not model-level, is that $\rho = 0.846$ across two very different models tells us the hardness is in the knowledge domain itself. You cannot train your way out of it.”

“We're honest about what we don't know: two models is suggestive not definitive, and we can describe which subjects are hard but not why. That's the natural next step.”

Land the practical implication last: “If you are deploying an instruction-tuned model on professional tasks — legal, medical, mathematical — the aggregate calibration number is not what you should be reporting. The gap is 13 times worse on some subjects than others, and that pattern holds across models.”

End confidently: “The framework is the contribution. It gives you a statistically honest answer